Completing the Square

Name _____

Algebra 2 Lesson 2-5 (fast track) Date _____ Period _____

Learning goal: I can solve quadratic equations by completing the square, even when the solutions are imaginary.

Language goal: I can explain each step of completing the square using the words perfect square trinomial, both sides, square root, and imaginary.

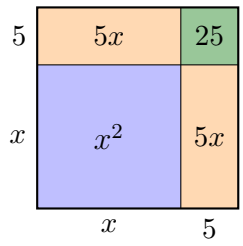
On the quiz: (1) Already a perfect square trinomial: take the square root of both sides (end of Example 3; full Savvas quiz only). (2) Bonus: write as $(x+p)^2 = q$ (first three steps of Example 3). (3) Solve by completing the square; the solutions are imaginary (Example 3, Try It 3). (4) A rectangle from its perimeter and area (full Savvas quiz only). (5) Zeros when x^2 has a coefficient: divide by it, then Example 3 (Practice).

Completing the square is finding the number to add to x^2+bx to make a **perfect square trinomial**. That number is **half the coefficient of the middle term, squared**:

$$x^2 + bx + \left(\tfrac{b}{2}\right)^2 = \left(x + \tfrac{b}{2}\right)^2$$

Example: x^2+10x. Half of 10 is 5, and $5^2 = 25$, so $x^2+10x+25 = (x+5)^2$. In an equation, add the number to **both sides** to keep it balanced.

Spanish: perfect square trinomial = *trinomio cuadrado perfecto*; completing the square = *completar el cuadrado*.



green 25 completes the square

Example 3: Solve a Quadratic Equation by Completing the Square

How can you solve $0 = x^2 - 2x + 3$ by completing the square?

$0 = x^2 - 2x + 3$	the equation
$-3 = x^2 - 2x$	move the constant: subtract 3 from both sides
$-3 + 1 = x^2 - 2x + 1$	$b = -2$: $\left(\frac{-2}{2}\right)^2 = 1$; add 1 to **both sides**
$-2 = (x-1)^2$	write the perfect square trinomial as $(x-1)^2$
$\pm\sqrt{-2} = x - 1$	take the **square root** of both sides ($\pm$)
$\pm i\sqrt{2} = x - 1$	$\sqrt{-2} = i\sqrt{2}$ (**imaginary**)
$1 \pm i\sqrt{2} = x$	add 1 to both sides

The solutions of $0 = x^2 - 2x + 3$ are $x = 1 + i\sqrt{2}$ and $x = 1 - i\sqrt{2}$.

Check $x = 1 + i\sqrt{2}$: $(1+i\sqrt{2})^2 - 2(1+i\sqrt{2}) + 3 = (-1 + 2i\sqrt{2}) - 2 - 2i\sqrt{2} + 3 = 0$ ✓

Say it: Half of -2, squared, is _____, so $x^2 - 2x + 1$ is a _____. I added it to both sides because … The square root of -2 makes the solutions imaginary because …

Try It 3: Solve by completing the square

Solve the following equations by completing the square.

a. $0 = x^2 + 4x + 8$

Half of b, squared: _____

$x =$ _____

b. $0 = x^2 - 8x + 17$

Half of b, squared: _____

$x =$ _____

In part a, why is there an i in the answer? Use the words *square root* and *imaginary*.

Practice: When x^2 has a coefficient

Find the zeros of the function $g(x) = -4x^2 + 8x - 5$ by completing the square.

Circle the coefficient of x^2. It is not 1, so first set $g(x) = 0$ and divide every term by it. Then complete the square as in Example 3.

$-4x^2 + 8x - 5 = 0$

Divide by _____: $x^2 - ____x + ____ = 0$

$x = \square + \square\, i$

$x = \square - \square\, i$

Why do the two zeros differ only in the sign of the i part?

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
FG7	Solve a quadratic equation using square roots (warm-up)	Quiz 1 (full Savvas quiz only); the square-root step of Example 3
9MW ★	Complete the square	Quiz 2 (bonus); the first steps of Quiz 3 and 5
NPH ★	Solve a quadratic equation by completing the square	Quiz 3 and 5; also 1 and 4 (full Savvas quiz only)
2YJ	Write a quadratic function in vertex form (optional)	optional: feeds no quiz item; vertex form from Lesson 2-1

SmartScore 70 or above on each starred skill replaces your quiz score.

Exit question: Construct Arguments (DOK 3)

Explain why you should not try to complete the square when solving $0 = x^2 - 4$.

Before the quiz, finish these sentences.

To complete the square for $x^2 + bx$, I add _____ to _____ of the equation.

The side with x is now a _____, so I can write it as $\left(x + \frac{b}{2}\right)^2$.

Then I take the _____ of both sides and remember the $\pm$.

If the number under the square root is negative, the solutions are _____ and have an i.